

Krzysztof Kotlarz

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Summary

I am a bioinformatics scientist and biostatistician with a PhD in biological sciences, working across genomics, genetics data analysis, statistics, and machine learning. My work spans high-throughput sequencing workflows, rare disease and clinical variant interpretation, multi-omics integration, population-scale genetic analyses, and explainable AI approaches for biological discovery.

Experience

Bioinformatician · *Ardigen* · Kraków, Poland

Jan 2025 – Present

- **Data Preparation & Multi-Omics Ingestion:** Architected robust high-dimensional multi-omics data ingestion and curation pipelines for genomic foundation models, enforcing strict metadata standards aligned with FAIR principles.
- **High-Throughput Variant Calling & Screen Optimisation:** Built high-throughput variant calling pipelines for rapid clinical and rare disease target discovery, alongside developing pipelines for precision siRNA off-target detection.
- **Nextflow Development:** Developed standardised, modular Nextflow pipelines to automate complex genomic and transcriptomic workflows, driving reproducibility and scalability across cross-functional drug discovery teams.
- **Reporting:** Built dynamic R Markdown templates to deliver decision-ready insights, integrating complex eQTL and regulatory data into concise technical reports.

Research Assistant · *Wrocław University of Environmental and Life Sciences* · Wrocław, Poland

Dec 2023 – Dec 2024

- **Teaching:** High-performance computing, Nextflow pipelines, Exploratory Data Analysis.
- **Projects:** Microbiome/metatranscriptome analysis.

Biostatistician · *Poznań University of Medical Sciences* · Poznań, Poland

Nov 2023 – Dec 2024

- **Variant Evaluation:** Execute WGS analysis for rare diseases, focusing on variant calling and interpretation to support molecular diagnostics.
- **Microarray Analysis:** Perform clinical and microarray data analysis to identify biomarkers and genetic patterns across patient cohorts.

Biostatistician (internship) · *Princess Margaret Cancer Centre* · Toronto, Canada

May 2023 – Oct 2024

- **Cancer Genomics:** Study of clinical biomarkers related to cancer patient outcomes, using biostatistics and machine learning algorithms.

Data Scientist · *Ryvü Therapeutics* · Wrocław, Poland

May 2023 – Oct 2024

- **Drug Discovery:** Statistical model design using high-performance methods related to drug discovery.
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Skills

Languages & Compute: Python, R, Nextflow, Bash

AI & Bio-AI: PyTorch, LLMOps, Agentic AI, Genomic Foundation Models

Omics & Analysis: WGS, Variant Calling, RNA-seq, GWAS, Illumina DRAGEN

Infra & Reproducibility: HPC, Docker, FAIR Data, Quarto

Education

PhD degree, Biological Sciences · *Wroclaw University of Environmental and Life Sciences* · Wroclaw, Poland
2020 – 2024

- The application of deep learning methods in the analysis of livestock genomes.

PhD degree, Biological Sciences · *Master's degree, Biostatistics and bioinformatics programming* · Wroclaw, Poland
2020 – 2022

- The comparison of high-performance methods in the variance components estimation process of mixed linear models.

PhD degree, Biological Sciences · *Wroclaw University of Environmental and Life Sciences* · Wroclaw, Poland
2017 – 2020

- The influence of selected factors on mortality among patients with acute heart failure.
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Projects

High-Sensitivity HPV Circulating Tumor DNA (ctDNA) Outperforms Conventional Imaging for Immediate Post-Treatment Risk Stratification in Oropharyngeal Cancer

International Journal of Radiation OncologyBiologyPhysics (2026) · [link](#)

- Evaluating ctDNA accuracy against conventional imaging for immediate post-treatment risk stratification.

Feature Selection Strategies for Deep Learning-Based Classification in Ultra-High-Dimensional Genomic Data

International Journal of Molecular Sciences (2025) · [link](#)

- Implementing and evaluating feature selection methodologies for deep learning models operating on high-dimensional genomic datasets.

Are head and neck versus abdominal paragangliomas driven by different single nucleotide events?

Journal of Applied Genetics (2025) · [link](#)

- Investigating distinct single nucleotide variant (SNV) profiles driving head/neck vs. abdominal paragangliomas.

Exploring the impact of sequence context on errors in SNP genotype calling with Whole Genome Sequence data using AI-based autoencoder approach

NAR Genomics and Bioinformatics (2024) · [link](#)

- Investigating the impact of sequence context on SNP genotype calling errors in Whole Genome Sequencing data using an AI-based autoencoder approach to improve accuracy.

Nextflow vs. plain Bash: Different Approaches to the Parallelisation of SNP Calling from the Whole Genome Sequence Data

NAR Genomics and Bioinformatics (2024) · [link](#)

- Comparing Nextflow and plain Bash approaches for parallelising SNP calling from whole-genome sequence data to evaluate their performance and efficiency.

An Explainable Deep Learning Classifier of Bovine Mastitis Based on Whole-Genome Sequence Data—Circumventing the p >> n Problem

International Journal of Molecular Sciences (2024) · [link](#)

- Designing an explainable deep learning classifier to identify bovine mastitis from whole-genome sequence data, effectively circumventing the "p >> n" problem.

IVIg Use Associated with Ten-Fold Reduction of Serious Infections in Multiple Myeloma Patients Treated with Anti-BCMA Bispecific Antibodies

Blood Cancer Discovery (2023) · [link](#)

- Analysis based on Self-Controlled Case Series (SCCS) model in patients with multiple myeloma using BCMA BiAb antibody treatment.

Better safe than sorry—Whole-genome sequencing indicates that missense variants are significant in susceptibility to COVID-19

PLoS ONE (2023) · [link](#)

- Evaluating functional missense variant burdens across whole-genome sequencing cohorts to assess COVID-19 susceptibility.

Application of mixed linear models for the estimation of functional effects on bovine stature based on SNP summary statistics from a whole-genome association study

Genetics Selection Evolution (2022) · [link](#)

- Comparison of high-performance methods for estimating variance components of linear mixed models.

Gene Variants Related to Cardiovascular and Pulmonary Diseases May Correlate with Severe Outcome of COVID-19

International Journal of Molecular Sciences (2022) · [link](#)

- Interpreting and analysing data from GWAS based on WGS data.

Beyond GWAS—could genetic differentiation within the allograft rejection pathway shape natural immunity to COVID-19?

International Journal of Molecular Sciences (2022) · [link](#)

- Constructing and applying a handwritten HPC algorithm to Linear Mixed Models using WGS data.

Identification of functional features underlying heat stress response in Sprague-Dawley rats using mixed linear models

Scientific Reports (2022) · [link](#)

- Constructing and applying a hand-written HPC algorithm in the application of Linear Mixed Models using gene expression data from rats.

The application of deep learning for the classification of correct and incorrect SNP genotypes from whole-genome DNA sequencing pipelines

Journal of Applied Genetics (2020) · [link](#)

- Designing and applying Deep Learning models in rare-event data using SNP data from WGS.

Achievements

Conference Scholarship

World Congress on Genetics Applied to Livestock Production (WCGALP 2022) · Jul 2022

- Awarded travel scholarship for presentation at the premier international livestock genetics conference.

Minister of Education and Science's Scholarship

Ministry of Education and Science, Poland · Oct 2021

- Prestigious national award granted for outstanding scientific achievements.

PRELUDIUM BIS 1 Research Grant

National Science Centre (NCN) · Oct 2020 – Sep 2024

- The application of deep learning methods in the analysis of livestock genomes* (~532,800 PLN / €125,000).
Principal Investigator / PhD Fellow for deep learning-driven genomic frameworks.

Chairman, Student Bioinformatic Scientific Group (SKN Bioinformatyków)

Wrocław University of Environmental and Life Sciences · Oct 2021 – Jun 2022

- Led academic initiatives, workshops, and scientific operations for the student bioinformatics association.
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Conferences

International Conference of Quantitative Genetics (ICQG 2024)

- *Vienna, Austria · Poster Presentation · Jul 2024*

European Society of Human Genetics (ESHG 2024)

- *Berlin, Germany · Poster Presentation · Jun 2024*

European Federation of Animal Science (EAAP 2023)

- *Lyon, France · Poster Presentation · Sep 2023*

European Federation of Animal Science (EAAP 2022)

- *Porto, Portugal · Oral Presentation · Sep 2022*

World Congress on Genetics Applied to Livestock Production (WCGALP 2022)

- *Rotterdam, Netherlands · Oral Presentation · Jul 2022*

European Federation of Animal Science (EAAP 2021)

- *Davos, Switzerland · Oral Presentation · Aug 2021*

European Federation of Animal Science (EAAP 2020)

- *Online · Oral Presentation · Nov 2020*

Polish Bioinformatics Society (PTBI 2020)

- *Online · Oral Presentation · Nov 2020*
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Courses

- **LARGE LANGUAGE MODELS ON SUPERCOMPUTERS** · VSC - Vienna Scientific Cluster · Sep 2024
- **MPI FOR PYTHON** · VSC - Vienna Scientific Cluster · Sep 2024
- **PYTHON FOR HPC** · VSC - Vienna Scientific Cluster · May 2024
- **FROM MACHINE LEARNING TO DEEP LEARNING: A CONCISE INTRODUCTION** · High-Performance Computing Center Stuttgart · Jun 2023
- **GOOGLE CLOUD FOR DATA ANALYST** · DaftAcademy · Dec 2022
- **KRZYSZTOF MĘDRELA WORKSHOPS PYTHON & MACHINE LEARNING** · Krzysztof Mędreła · Nov 2020